

$\mathbb{RP}^3 \times \mathbb{R}$ as a Limiting Case of T^4 :

**Time Compactification, Cosmological Scale, and
the Embedding
of the σ -Programme within FFGFT Geometry**

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Abstract

The σ -programme of Kriger (2026) postulates the cosmological topology $RP^3 = S^3/Z_2$ with an open time dimension ($RP^3 \times \mathbb{R}$). The Fundamental Fractal-Geometric Field Theory (FFGFT) postulates a 4D identification torus $T^4 = (S^1)^4$ in which all four dimensions including time are compactified, with the foundational relation $\tilde{T} \cdot m = 1$. This document shows that $RP^3 \times \mathbb{R}$ is the limiting case of T^4 for vanishing mass ($m \rightarrow 0$): the time compactification radius $R_t = \hbar/(mc^2)$ diverges for small masses, so the time dimension appears effectively continuous at cosmological scales. The σ -programme thereby correctly describes the low-energy limit of the T^4 geometry, but requires two external suppression mechanisms for the cosmological constant because the temporal contribution to topological suppression is absent. We work in natural units with $c = \hbar = k_B = 1$ unless stated otherwise.

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.1 Starting Point: Two Topological Frameworks

Two independently developed frameworks postulate a compact cosmological topology beyond the Standard Model:

The σ -programme (Kriger 2026) [1]. Cosmological topology $RP^3 = S^3/Z_2$ — a compact antipodal three-dimensional manifold. The time dimension remains open: spacetime is $RP^3 \times \mathbb{R}$. Three fundamental problems dissolve geometrically: the Yang-Mills mass gap, baryon asymmetry, and the cosmological constant.

FFGFT [2]. 4D identification torus $T^4 = (S^1)^4$ — all four dimensions compactified. The foundational relation $\tilde{T} \cdot m = 1$ couples time compactification and mass dually. From a single parameter $\xi = 4/3 \times 10^{-4}$ all Standard Model constants follow.

The central question: are these frameworks competing or mutually embeddable?

.2 Time Compactification in FFGFT

In FFGFT the time dimension is compactified with radius

$$R_t = \frac{\hbar}{mc^2}. \quad (.1)$$

This radius depends on the mass of the system under consideration and takes characteristic values at different physical scales:

System	Mass	R_t	Comparison
Electron	$9.1 \times 10^{-31} \text{ kg}$	$3.9 \times 10^{-13} \text{ m}$	$\ll R_U$
Proton	$1.7 \times 10^{-27} \text{ kg}$	$2.1 \times 10^{-16} \text{ m}$	$\ll R_U$
Higgs boson	$2.2 \times 10^{-25} \text{ kg}$	$1.6 \times 10^{-18} \text{ m}$	$\ll R_U$
Λ -scale	$m_\Lambda \sim \hbar H_0 / c^2$	$\sim 10^{38} \text{ m}$	$\gg R_U$

For the cosmological energy scale (associated with H_0):

$$m_\Lambda = \frac{\hbar H_0}{c^2} \approx 2.5 \times 10^{-69} \text{ kg} \quad \Rightarrow \quad R_t^\Lambda \approx \frac{\hbar}{m_\Lambda c^2} \cdot c \approx 10^{38} \text{ m} \gg R_U. \quad (.2)$$

The time compactification at cosmological scales exceeds the Hubble radius by **12 orders of magnitude**.

.3 $\mathbb{RP}^3 \times \mathbb{R}$ as a Limiting Case of T^4

The Limiting Transition

Let $T^4 = S_{R_1}^1 \times S_{R_2}^1 \times S_{R_3}^1 \times S_{R_t}^1$ be the 4D torus with three spatial radii $R_1, R_2, R_3 \sim R_U$ and time radius $R_t = \hbar/(mc^2)$.

In the limit $m \rightarrow 0$:

$$R_t = \frac{\hbar}{mc^2} \xrightarrow{m \rightarrow 0} \infty. \quad (.3)$$

A circle S_R^1 with $R \rightarrow \infty$ becomes topologically the real line $\mathbb{R}$. The torus degenerates:

$$T^4 \xrightarrow{R_t \rightarrow \infty} S_{R_1}^1 \times S_{R_2}^1 \times S_{R_3}^1 \times \mathbb{R} \cong T^3 \times \mathbb{R}. \quad (.4)$$

The spatial component T^3 is not identical to $\mathbb{RP}^3$; the antipodal identification Z_2 must be added. In T^4 this corresponds to the symmetry $\tilde{T} \rightarrow -\tilde{T}$, i.e. time reversal.

$\mathbb{RP}^3 \times \mathbb{R}$ is the limiting case of T^4 for $m \rightarrow 0$, with the antipodal time-reversal symmetry preserved as residual structure.

Physical Interpretation

The time compactification with $R_t \sim 10^{38}$ m is completely invisible on the scale of cosmological observations ($R_U \sim 4.4 \times 10^{26}$ m):

$$\frac{R_t^\Lambda}{R_U} \approx 10^{11}. \quad (.5)$$

Every observer at cosmological scales perceives the time dimension as effectively open. The σ -programme correctly describes this observational reality.

.4 Mode Spectrum and UV Behaviour

The structural difference appears in the mode spectrum:

$\mathbf{RP}^3 \times \mathbb{R}$: The time momentum k runs continuously from $-\infty$ to $+\infty$. The zero-point energy contains the integral:

$$E_{\text{ZP}} \sim \int_0^{k_{\text{Pl}}} dk \sum_{\ell \text{ even}} (\ell + 1)^2 \sqrt{k^2 + \frac{\ell(\ell + 2)}{R_U^2}}. \quad (.6)$$

This integral diverges as $k_{\text{Pl}}^4 \sim (1/l_P)^4$, yielding the familiar 10^{113} J/m³. Kriger's antipodal cancellation (Stage 1) suppresses this divergence.

T^4 : The time momentum is discrete: $k_{t,n} = n/R_t$, $n \in \mathbb{Z}$. The zero-point energy is:

$$E_{\text{ZP}} \sim \sum_{\mathbf{n} \in \mathbb{Z}^4} \sqrt{\frac{n_1^2}{R_1^2} + \frac{n_2^2}{R_2^2} + \frac{n_3^2}{R_3^2} + \frac{n_t^2}{R_t^2}}. \quad (.7)$$

This sum is finitely regularisable via the spectral zeta function on T^4 *without an external cutoff*. No UV problem arises.

Consequence: The σ -programme requires two external suppression mechanisms (antipodal cancellation + α LGQV sequestering) because time compactification is absent. In T^4 , the compactified time dimension supplies this suppression structurally.

.5 Comparison of Cosmological Constant Derivations

Aspect	σ -programme ($\mathbb{RP}^3 \times \mathbb{R}$)	FFGFT (T^4)
Time dimension	Open ($\mathbb{R}$)	Compactified ($R_t = \hbar/mc^2$)
UV divergence	Present, needs Stage 1	Structurally absent
Stage 1	Antipodal cancellation (10^{-77})	Not required
Stage 2	α LGQV sequestering (10^{-46})	Not required
Exponent n	≈ 1.1 , not exactly derived	Not applicable
Final formula	$\rho_\Lambda \sim (R_{\text{had}}/R_U)^n \Lambda_{\text{QCD}}^4$	$\hbar H_0/m_e c^2 = (\pi/2)\xi^{10}$
Deviation	Order-of-magnitude correct	0.75%
Free parameters	Exponent n	None

.6 Domains of Validity

The σ -programme is valid:

- At cosmological scales ($R \sim R_U$) where $R_t \gg R_U$
- For systems where time compactification is invisible
- As an effective description of the T^4 geometry at $m \rightarrow 0$
- For qualitative topological statements on baryon asymmetry, Yang-Mills, and Λ

FFGFT is required:

- At particle scales where $R_t \ll R_U$ and time compactification is physically relevant
- For quantitative derivations of masses, α , G
- Wherever winding numbers in the time dimension determine quantum numbers
- As the complete 4D theory that contains $RP^3 \times \mathbb{R}$ as a limiting case

.7 Summary

1. $RP^3 \times \mathbb{R}$ is the limiting case of T^4 for $m \rightarrow 0$, i.e. at scales where the time compactification $R_t = \hbar/(mc^2)$ far exceeds the Hubble radius.
2. At cosmological scales, $R_t \sim 10^{38} \text{ m} \gg R_U$: the time dimension appears effectively open.
3. The σ -programme correctly describes the observational reality at these scales — as the low-energy limit of T^4 .
4. The two suppression mechanisms of the σ -programme (antipodal cancellation + α LGQV) implicitly compensate for the missing temporal contribution to topological suppression.
5. In T^4 , the compactified time dimension supplies this contribution structurally; the cosmological constant follows directly from ξ without external mechanisms.
6. The frameworks are therefore not competing but hierarchically embedded: σ -programme $\subset$ FFGFT in the limit $m \rightarrow 0$.

Methodological classification: This comparison constitutes a plausibility bridge. The formal embedding $RP^3 \subset T^4$ via antipodal time reversal requires further mathematical elaboration. The numerical statements (table in Section 2) are directly verifiable.

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Bibliography

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